

A Multi-Stage Framework for Automated Document Understanding Using Deep Learning and Large Language Models

Final report

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ABSTRACT The automatic analysis of visual documents that exhibit diverse content and layout characteristics remains an open and challenging problem for document understanding systems. While traditional Optical Character Recognition (OCR)-based approaches are effective at extracting text, they operate without considering document type or contextual information, which limits their semantic understanding capabilities. This study presents a multi-stage and modular document understanding system that aims to analyze visual documents not only at the textual level but also at the content and contextual levels. At the core of the proposed system lies a deep learning-based document classification module, which provides essential contextual information for subsequent analysis stages. Following the document classification step, OCR-based text extraction is performed, and the extracted text is processed by a Large Language Model (LLM)-based analysis component. Through this design, the system is able to handle different types of documents in an end-to-end manner and automatically structure their core content information. The proposed approach is evaluated on a subset of the RVL-CDIP dataset, and the role of the document classification module within the overall document understanding process is analyzed.

INDEX TERMS Document Understanding, Deep Learning, Document Classification, Large Language Models (LLMs), Optical Character Recognition (OCR), Information Extraction.

I. INTRODUCTION

Visual documents remain one of the most common carriers of information in both corporate and academic environments. Documents such as reports, forms, official correspondence, and similar materials exhibit diverse content types and layout structures without adhering to a fixed schema, which poses significant challenges for automated document understanding systems. In particular, the manual inspection of such unstructured or semi-structured visual documents is not only time-consuming but also prone to human error.

Optical Character Recognition (OCR)-based approaches are widely adopted as a fundamental component for extracting text from visual documents. However, OCR outputs do not directly provide information about document type, contextual meaning, or structural characteristics. Moreover, the extracted text is often lengthy and dense, making it difficult for users to directly access the specific information they require. As a result, analyzing raw OCR

text in isolation becomes challenging, and distinguishing between different document types at a semantic level remains a non-trivial task.

In recent years, deep learning-based models have achieved significant progress in the field of visual document classification. By leveraging visual features, these models are capable of predicting document types with high accuracy and providing strong contextual cues about document content. In this context, document classification has evolved beyond a simple labeling task and has become a critical component that enables subsequent analysis stages to be conducted in a more consistent and targeted manner.

This study presents a multi-stage and modular document understanding system designed to perform end-to-end analysis of visual documents. In the proposed approach, a deep learning-based document classification module is positioned at the core of the system, supplying essential document-type information that serves as contextual grounding for OCR-based text extraction and Large

Language Model (LLM)-supported content analysis. Through this design, the system is able to process documents in a type-aware manner without requiring additional manual inspection or pre-classification by the user.

The objective of this approach is to move beyond traditional OCR-based systems by enabling more meaningful, consistent, and structured information extraction from visual documents. The proposed system is evaluated on a subset of the RVL-CDIP dataset, and the role of the document classification module within the overall document understanding process is analyzed..

II. Literature – Background Info – Related Work

Document understanding has been an active research area at the intersection of computer vision and natural language processing. Early studies demonstrated that visual characteristics alone can provide strong signals for document categorization. Harley et al. [1] systematically evaluated deep convolutional neural networks for document image classification and retrieval, establishing RVL-CDIP as a standard benchmark and showing that deep visual models can effectively distinguish document types based on layout and appearance.

Building upon document classification, subsequent research has focused on holistic document understanding by jointly modeling textual content and layout information. Xu et al. [2] introduced LayoutLM, which integrates text, spatial layout, and visual features, achieving significant improvements across multiple document understanding tasks. In a similar direction, Kim et al. [3] proposed an OCR-free document understanding transformer that directly operates on document images, highlighting the importance of visual and structural cues beyond plain text extraction.

More recent advances have explored the integration of large language models and multimodal reasoning capabilities. Yao et al. [4] presented ReAct, a framework that combines reasoning and acting in language models, illustrating how modular and agent-inspired designs can improve complex decision-making processes. In parallel, BLIP-2 [5] have advanced multimodal learning by aligning visual encoders with large language models, enabling richer cross-modal understanding and reasoning over visual inputs.

Despite these advances, many existing approaches implicitly rely on models to infer document context without explicitly incorporating document type as a guiding signal. In contrast, this work positions deep learning-based document classification as a central and explicit component of the document understanding pipeline. By leveraging document type information as contextual grounding for downstream OCR-based extraction and LLM-supported analysis, the proposed system aligns with prior multimodal research while emphasizing the role of classification in enabling more structured and consistent document understanding.

III. Methodology

A. System Architecture Overview

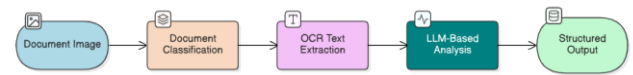


Figure 1: System Architecture Diagram

This study proposes a multi-stage and modular document understanding architecture designed to perform end-to-end analysis of visual documents. The system is structured as a linear processing pipeline that takes a document image as input and sequentially applies document classification, OCR-based text extraction, and Large Language Model (LLM)-supported content analysis. An overview of the proposed system architecture is illustrated in Fig. 1.

In the first stage of the pipeline, the input document image is processed by a deep learning-based document classification module. This module leverages visual features of the document to identify its type, providing essential contextual information for subsequent processing stages. The document type information serves as a guiding signal that enables later analysis steps to be performed in a more consistent and targeted manner.

Following the classification stage, the document image undergoes OCR-based text extraction. In this step, the visual content of the document is converted into digital text, producing a raw textual representation. The OCR process is applied uniformly across all document types and functions as a fundamental component responsible for text digitization, independent of document-specific assumptions.

In the final stage, the extracted OCR text is processed by an LLM-based analysis component. This component utilizes the document type information obtained from the classification module as contextual guidance to perform higher-level content analysis and to generate structured outputs. Through this design, the system goes beyond simple text extraction and aims to deliver meaningful, organized information that directly supports user needs.

B. Document Classification Module

The document classification module constitutes the core deep learning component of the proposed document understanding pipeline. The primary objective of this module is to identify the type of an input document image and to provide high-level contextual information that guides subsequent processing stages. Given the heterogeneous nature of visual documents in the RVL-CDIP dataset, explicitly determining the document type enables a type-aware analysis process by preventing all documents from being treated as a uniform text block.

To implement this module, a transfer learning approach is adopted using pre-trained visual backbone architectures. Within the scope of this study, EfficientNet-B0 and Swin Transformer architectures are evaluated and compared. Both models are fine-tuned on a selected subset of the RVL-CDIP dataset, allowing general-purpose visual representations to be

adapted to document-specific characteristics such as layout, typography, and structural patterns.

EfficientNet-B0 is considered as a baseline convolutional neural network architecture due to its balanced trade-off between model complexity and performance. In contrast, the Swin Transformer architecture offers a hierarchical transformer-based design capable of modeling both local and global visual dependencies through its shifted window attention mechanism. This property makes Swin Transformer particularly well-suited for document images, where layout and structural relationships play a critical role.

During training, document images are resized and normalized to ensure compatibility with the selected backbone architectures. The classification task is formulated as a multi-class image classification problem, in which each document image is associated with a single document type label. The output of the classification module is the predicted document category, which is subsequently utilized as contextual input for downstream processing stages.

By explicitly separating the document classification stage from OCR-based text extraction and LLM-supported content analysis, the proposed system enforces a modular design. This design choice allows document type information to be computed once and reused across subsequent stages, thereby establishing the document classification module as a foundational building block of the overall document understanding process.

C. OCR-Based Text Extraction

OCR-based text extraction is positioned as a core supporting component within the proposed document understanding pipeline, responsible for converting visual document content into digital text. The primary objective of this module is to extract the textual information present in document images as accurately and comprehensively as possible, thereby providing the necessary textual input for subsequent content analysis stages.

In this study, the Chandra model is employed for OCR processing. Chandra is an OCR model developed using a vision–language pretraining approach and is specifically optimized for text extraction from document images. By leveraging contextual understanding of visual and textual cues, the model is capable of producing more consistent and semantically coherent text outputs compared to traditional OCR systems that rely solely on character-level recognition.

Within the proposed system, the OCR process is applied uniformly across all document types, independent of the predicted document category. This design choice ensures that the OCR module remains solely responsible for text digitization without incorporating document-type-specific assumptions. Document type information obtained from the classification module is intentionally excluded from the OCR stage and is instead utilized in downstream analysis components.

The OCR output typically consists of long and dense textual content. For this reason, the extracted raw text is not directly presented to the user but is forwarded to the LLM-based analysis stage. This approach allows the system to move beyond simple text digitization and enables higher-level interpretation and structuring of document content in alignment with user needs.

By explicitly separating OCR-based text extraction from document classification and LLM-supported analysis, the proposed architecture maintains a modular design. This separation clarifies the role of each component and contributes to a more flexible, interpretable, and extensible end-to-end document understanding system.

D. LLM-BASED DOCUMENT ANALYSIS

In this study, the Large Language Model (LLM) is employed to enable a more meaningful and goal-oriented analysis of textual content, rather than directly presenting raw OCR outputs to the user. The LLM component is not designed as an independent reasoning engine; instead, it operates as a controlled analysis module that is explicitly guided by the outputs of the document classification stage.

The LLM-based analysis process is divided into two distinct operating modes based on the document type prediction and its associated confidence score produced by the classification module. When the classification confidence exceeds a predefined threshold, the LLM is activated in a document-type-specific analysis mode. In this mode, the model performs targeted information extraction based on a predefined set of fields relevant to the predicted document category. The LLM is strictly instructed to generate outputs in JSON format and to include only information that is explicitly present in the OCR text, while omitting any fields that cannot be reliably identified.

Conversely, when the classification confidence is low or the document type cannot be determined with sufficient reliability, the system transitions to a generic analysis mode to avoid type-based assumptions. In this mode, the LLM is limited to producing a concise summary of the document content, identifying its main topic and highlighting notable information elements. This design choice prevents the generation of erroneous or assumption-driven structured outputs under uncertainty.

To further enhance output reliability, the LLM is constrained to produce only valid JSON responses. Any output containing explanatory text, formatting, or invalid structure is rejected by the system. Additionally, excessively long OCR outputs are truncated prior to analysis to reduce the risk of context drift and inconsistent model behavior.

Through this design, the LLM becomes a context-aware and controlled analysis component that is explicitly directed by the document classification module, which serves as the central decision-making element of the pipeline. The proposed approach offers a balanced trade-off between flexibility and reliability in LLM-based document analysis,

particularly in the presence of heterogeneous and partially structured document data.

IV. Experiments

A. DATASET

All experiments in this study were conducted on a selected subset of the **RVL-CDIP** dataset, a widely used benchmark for document image classification. The subset consists of **email, invoice, letter, and form** classes, which represent common document types encountered in corporate environments. The dataset was divided into training, validation, and test splits.

B. DOCUMENT CLASSIFICATION EXPERIMENTS

The document classification module was trained using a transfer learning approach. Two architectures were evaluated and compared: **EfficientNet-B0** and **Swin Transformer**. EfficientNet-B0 was selected as a baseline model due to its lightweight structure and computational efficiency, while Swin Transformer was examined for its ability to better capture layout and structural relationships in document images.

For EfficientNet-B0, experiments were conducted to analyze the effects of dataset size and input resolution on classification performance. Across all experiments, the learning rate was fixed at **2e-4**, batch size was set to **32**, and training was performed for **15 epochs**. Results show that increasing input resolution alone does not improve performance; instead, using a resolution compatible with the model architecture (**224 × 224**) combined with increased data volume leads to more stable learning. Under this configuration, a validation accuracy of **90.0%** was achieved.

When trained on the same dataset, the Swin Transformer model consistently outperformed EfficientNet-B0 on both validation and test sets, demonstrating the effectiveness of transformer-based architectures for document image classification.

C. OCR MODEL SELECTION

During the OCR stage, multiple OCR models were empirically evaluated in terms of text extraction quality and computational efficiency. Lightweight models such as **DeepSeek OCR** offer fast inference times but produce unstable outputs for certain document types. These issues include repeated words and degraded textual coherence. Although such errors do not constitute hallucinations in a strict sense, they negatively affect downstream analysis.

In contrast, **HunyuanOCR** produces more stable and cleaner text outputs but suffers from higher inference latency, making it less suitable for practical deployment. Therefore, **Chandra OCR** was selected as a balanced solution between quality and efficiency. Chandra generates coherent and structurally consistent text outputs while maintaining

reasonable processing times, making it suitable for integration into the proposed multi-stage document understanding pipeline.

D. TRAINING ENVIRONMENT AND HARDWARE

All document classification experiments were conducted using transfer learning on a standard Google Colab T4 GPU, and the training process was completed within a short time frame. The end-to-end pipeline, including OCR inference with the Chandra OCR model and LLM-based analysis, was executed separately on an NVIDIA A100 GPU due to the higher computational requirements of vision-language models. No specialized hardware was required beyond GPU acceleration for inference.

V. Results & Analysis

This section presents both quantitative and qualitative evaluations of the proposed document understanding system. Given the multi-stage nature of the pipeline, different evaluation strategies are applied to each component. While the document classification module is assessed using numerical metrics, the OCR and LLM components are evaluated through qualitative analysis.

A. DOCUMENT CLASSIFICATION PERFORMANCE

The performance of the document classification module was evaluated on validation and test sets. Within the scope of this study, two deep learning architectures were comparatively analyzed: **EfficientNet-B0** and **Swin Transformer**.

The EfficientNet-B0 model achieved **90.0% validation accuracy** and **88.7% test accuracy**. The training and validation accuracy and loss curves for EfficientNet-B0 are presented in **Figure 2**.

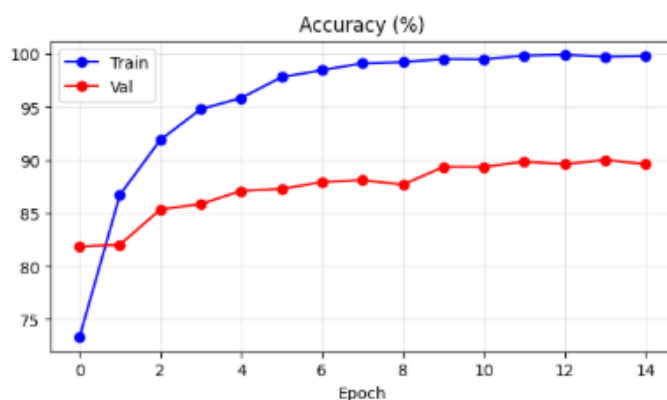


Figure 2: EfficientNet-B0 Training & Validation Accuracy

Analysis of the learning curves shows that EfficientNet-B0 converges rapidly; however, a noticeable gap between training and validation accuracy is observed, indicating mild overfitting due to the limited dataset size.

The primary classification model, **Swin Transformer**, outperformed EfficientNet-B0, achieving **92.0% validation accuracy** and **91.3% test accuracy**. The corresponding learning curves are shown in **Figure 3**.

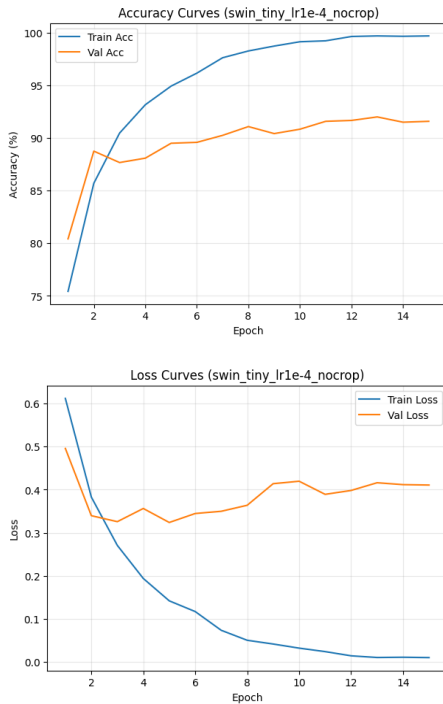


Figure 3: Swin Transformer Training & Validation Accuracy/Loss
Compared to EfficientNet-B0, Swin Transformer exhibits more stable convergence and a smaller gap between training and validation performance. This behavior indicates improved generalization and highlights the effectiveness of transformer-based architectures in modeling layout and structural dependencies in document images. Overall, these results demonstrate that Swin Transformer provides superior performance for document image classification tasks, particularly in scenarios where spatial organization and global contextual information are critical. To further analyze class-wise performance, a confusion matrix generated for the primary classification model, **Swin Transformer**, is presented in **Figure 4**.

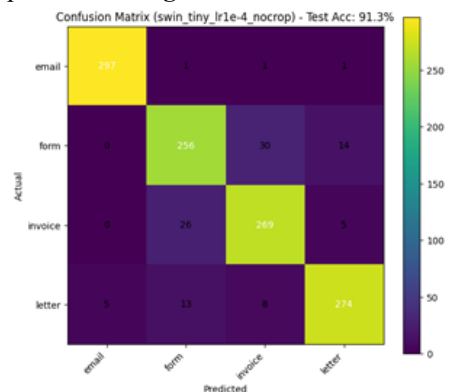


Figure 4: Confusion Matrix (Swin Transformer)

The confusion matrix indicates that the **email** and **letter** classes are classified with high accuracy. These document types exhibit distinctive visual characteristics in terms of layout structure, heading patterns, and text density, which enables the model to learn robust class representations.

In contrast, limited confusion is observed between the **invoice** and **form** classes. This confusion arises from similarities in visual layout, including structured fields, tabular regions, and form-like content blocks commonly shared by both document types. Such structural overlap makes fine-grained visual discrimination more challenging. These findings highlight the impact of inter-class visual similarity on document classification performance and further justify the role of document classification as a contextual guidance mechanism for downstream document analysis stages.

B. QUALITATIVE EVALUATION OF OCR AND LLM COMPONENTS

Due to the absence of ground-truth structured annotations for OCR and LLM outputs, these components are evaluated qualitatively rather than through numerical accuracy metrics. OCR model comparison reveals that lightweight models such as DeepSeek OCR produce fast outputs but occasionally suffer from repeated words and degraded textual coherence. In contrast, heavier models like HunyuanOCR provide more stable outputs at the cost of significantly higher inference latency. The selected Chandra OCR model offers a balanced trade-off between output quality and computational efficiency.

The LLM component is evaluated based on behavioral consistency rather than extraction accuracy. The model operates in a controlled manner, producing structured outputs only when classification confidence is high and switching to a generic summarization mode under uncertainty. This behavior reduces the risk of assumption-driven or incorrect structured outputs.

C. END-TO-END PIPELINE BEHAVIOR

The end-to-end behavior of the proposed system is evaluated in terms of decision consistency rather than a single end-to-end accuracy metric, due to the modular and multi-stage nature of the pipeline.

The document type and confidence score produced by the classification module serve as the primary mechanism for controlling downstream components. **Figure 5** provides a visual overview of the confidence-based routing behavior within the system.

When the classification confidence is high, the system consistently activates type-specific information extraction. In contrast, low-confidence predictions trigger a generic and safer analysis mode that avoids type-based assumptions. This design choice enables the system to operate more conservatively under uncertainty, preventing incorrect or

assumption-driven structured outputs. Overall, the results demonstrate that the proposed document understanding pipeline exhibits predictable, stable, and context-aware behavior.

```

Dosya: 2078123093a.tif
=====
ADIM 1: Classification
-----
The following generation flags are not valid and may be ignored:
Tahmin: email
Güven: 100.0%

ADIM 2: OCR (Chandra)
-----
Çıkarılan: 216 karakter

ADIM 3: LLM Analiz
-----
✓ Yüksek güven (100.0%) → EMAIL analizi
Mod: type_specific

=====
✓ İşlem tamamlandı!
=====

SONUÇ RAPORU
=====

SINIFLANDIRMA:
Tür: email
Güven: 100.0%
Mod: ✓ Türe Özgü Analiz

OCR ÇIKTISI:
From: Lindheim, Jim(JL) on Sat, Jan 27, 1996 6:53 AM
Subject: Charlotte Sessions
To: Firesione, Marc; Purcell, Clare; Winokur, Matt
File(s): ETS/Charlotte Sessions

Please see attached memo on the Charlotte Sessions.

LLM ANALİZ SONUCU:
-----
{
  "from": "Lindheim, Jim(JL)",
  "to": "Firesione, Marc; Purcell, Clare; Winokur, Matt",
  "date": "Sat, Jan 27, 1996 6:53 AM",
  "subject": "Charlotte Sessions"
}
=====

```

Figure 5: End-to-End Pipeline Behavior (Routing Example)

VI. Conclusion

This study presented a modular and multi-stage framework for automated document understanding, designed to incorporate document-level context into downstream analysis rather than relying solely on text extraction. Unlike traditional OCR-based pipelines, the proposed system leverages document classification as a central component to guide subsequent OCR and LLM-based processing stages. Experimental results demonstrate that transformer-based architectures, particularly Swin Transformer, outperform convolutional models in document image classification tasks by more effectively capturing layout and structural dependencies. The classification module functions not only as a labeling mechanism but also as a decision-making

component that controls the behavior of downstream OCR and LLM modules.

Despite these contributions, several limitations remain. The absence of structured ground-truth annotations for OCR and LLM outputs restricts the use of quantitative evaluation metrics for these components, necessitating qualitative analysis based on behavioral consistency. Additionally, experiments were conducted on a limited subset of the RVL-CDIP dataset, and the system was not evaluated across a broader range of document categories.

Future work will focus on expanding the number of supported document classes, constructing structured evaluation datasets for downstream components, and performing more comprehensive end-to-end system analysis. Furthermore, incorporating more advanced routing strategies and multi-step document reasoning capabilities may further enhance the robustness and applicability of the proposed document understanding framework.

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